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Integrating EEG and NIRS improves BCI performance during motor imagery *

Zhongpeng Wang[1], Cong Cao[2], Yijie Zhou[2], Long Chen[2], Bin Gu[1], Shuang Liu[2], Minpeng Xu[1], Feng He[1], Dong Ming[1, 2], *Senior Member, IEEE*

***Abstract*—Hybrid brain-signal could theoretically achieve a more accurate decoding in brain-computer interfaces (BCIs). This study aims to improve the BCI performance by integrating electroencephalographic and cerebral hemodynamic patterns. Sixteen volunteers participated in a seven-session motor imagery (MI) based visual-haptic neurofeedback training (NFT) experiment. Electroencephalogram (EEG) and near infrared spectroscopy (NIRS) signals were synchronously recorded during the transient NFT. Time-frequency and topology analysis demonstrated that repetitive and continuous visual-haptic NFT conditions could induce an enhancement on cortical activations. Additionally, a classifier calibration strategy was proposed by integrating both the EEG and NIRS information into a strong classifier. The results revealed that the classifier constructed on integrating EEG and NIRS patterns was significantly superior to that only with independent information, achieving ~14% and 5% improvement respectively and reaching 84.5% in mean classification accuracy. Therefore, integrating EEG and NIRS patterns is an effective classifier calibration strategy to improve the MI-BCI performance, and could also be applied to other BCI paradigms.**

I. Introduction

There is increasing interest in exploring the effect of neurofeedback training (NFT) as a therapeutic technique for a range of neurological diseases such as stroke, spinal cord injury (SCI) and amyotrophic lateral sclerosis (ALS) [1]. Neural plasticity is critical for brain function and has been extensively explored over the past years using non-invasive imaging, challenging our understanding of the brain's capacity to adapt in response to internal and external stimuli. Previous findings have been widely exploring current understanding of how fast the human brain change its structural and functional connections in response to training [2]. However, it is still unclear what degree the NFT could contribute to functional changes or enhancements.

Previous studies have shown that NFT can be used to downregulate arousal and improve human performance in a demanding sensory-motor task in real time. NFT was progenitor of brain–computer interfaces (BCIs) [3]. BCIs are starting to prove their efficacy as assistive and rehabilitative technologies in patients with severe motor impairments. Additionally, EEG-based BCIs have enabled some paralyzed patients to communicate, but near-infrared spectroscopy (NIRS) combined with a classical conditioning paradigm is the only successful approach for complete locked-in syndrome [4].

More generally, the joint analysis of different hybrid brain-signals can shed light on the complex relationships or connects between anatomical, functional and electrophysiological properties of the brain [5]. Integration of EEG and NIRS allows for an "augmented" analysis of the neuronal dynamics, as the single modalities provide a partial estimation of the underlying neural activity. Joint EEG-NIRS analyses fall in relatively many categories and methods. In the first, information extracted from one methodology are integrated or drive the analysis of the second while in symmetrical approaches (data fusion) a joint generative model is used, specifically, in BCIs [6]-[8]. These approaches have been poorly explored, complexity of neurovascular coupling being their main limitation.

In this paper, we proposed a method of integrating EEG and NIRS patterns to improve the BCI performance after a transient NFT. Firstly, in our previous work [9], we have demonstrated a BCI based visual-haptic NFT could enhance cortical activations and classification performance during motor imagery. Then, the Graz motor imagery based BCI (Graz MI-BCI) paradigm was utilized to explore the changes of BCI performance before and after NFT. Also, the EEG and NIRS signals were recorded synchronously. We compared different classification methods under different conditions of EEG and NIRS patterns to optimize BCI performance.

II. Methods

A. Participants

Sixteen right-handed adults (5 females, mean age: 23.2 ± 2.4, range21 to 28 years old) participated in this experiment. All subjects are healthy without any history of neurological illness or limb movement disorder and signed a consent form in advance. The procedure of the experiment was clearly explained to each subject before data recording. This study was approved by the ethical committee of Tianjin University.

B. Experimental Paradigm

Our previous work has proposed the detailed procedure of this experiment [9]. The experiment was divided into four task stages and consisted of seven sessions (one screening session, two previous control sessions, two neurofeedback training sessions and two posterior control sessions) in total. Fig. 1 illustrates the experimental paradigm procedure. Subjects could have an impermanent break between these sessions. For the first screening session, all the experimental scenarios

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[1] College of Precision Instruments & Optoelectronics Engineering, Tianjin University, Tianjin, 300072 China.

[2] Academy of Medical Engineering and Translational Medicine, Tianjin University, Tianjin, 300072 China.

would appear on the LCD screen. Subjects were asked to view and get familiar with the experimental paradigm. The screening session lasted about 5 minutes. The next two task sessions were the previous control sessions utilizing Graz MI-BCI paradigm [10]. And then, two visual-haptic neurofeedback training sessions were conducted, the visual feedback is shown in Fig. 1(c), there are four moving angles of virtual hand in the screen :0°,15°,30° and 45° for real-time feedback training feature parameter (real-time lrERD) level. The 30s baseline datas are used to calculate the feedback threshold(lrERD). When the training feature parameter lower than the threshold, the virtual hand will away from the needle. On the contrary, when the training feature parameter higher than the threshold, the virtual hand will touch the needle and the bilateral palm of the subject will be given an electrical stimulation, Fig. 1(d) illustrates the electrical stimulation demonstration and setup. Subjects were asked to apply the kinesthetic MI strategy to let the virtual hand away from the needle. At last, there were two posterior control sessions same as the previous control sessions.

C. EEG and NIRS data acquisition and preprocessing

As shown in Fig. 2(a), subjects were seated comfortably in a chair about 1 m away from a 23.5-inch LCD screen. EEG signals were acquired by a 64-channel SynAmps2 system (Neuroscan, Australia) with standard Ag/AgCl electrodes placed on the scalp according to the international 10-20 system. The reference electrode was placed on the nose and the ground electrode was placed on the forehead. The impedance for all electrodes was kept below 10 kΩ. The EEG signals were sampled at 1000 Hz. An online band-pass filter between 0.5 and 100 Hz and 50 Hz notch filter were enabled in the amplifier to filter the high-frequency noise, baseline drift and power line interference. For the preprocessing, the EEG raw data were downsampled at 200 Hz, then processed by the common average reference (CAR). A multi-channel NIRS system (NirScan, Danyang Huichuang Medical Equipment Co., Ltd., China) was utilized for cerebral blood oxygen acquisition. The NIRS signals were sampled at 10 Hz. An online band-pass filter between 0.01 and 3 Hz.

NIRS channels cover the left and right sensorimotor brain regions (corresponding to left and right limb motor imagery evoked cortex), which are the regions of interest (ROI) for data analysis of cerebral blood oxygen. Time course analysis (TCA) follows the coherence averaging method of event-related potentials, i.e. taking event labels as criteria to intercept valid time data and characterize concentration change (CC) of [oxy-Hb] and [deoxy-Hb].

D. NFT system design and online EEG processing

Here, the feedback threshold was set to the mean lateralized relative ERD (lrERD) power in the eyes-open relaxing baseline for LH-MI or RH-MI training respectively at α (8-13Hz) and β (15-29Hz) rhythm band. The NFT parameter for the LH-MI or RH-MI training was the lateralized relative ERD power between C3 and C4 channels, which was calculated by the following equation:

For RH-MI training parameter:

$$lrERD_{C3-C4} = RP_{ERD,C3} - RP_{ERD,C4} \tag{1}$$

For LH-MI training parameter:

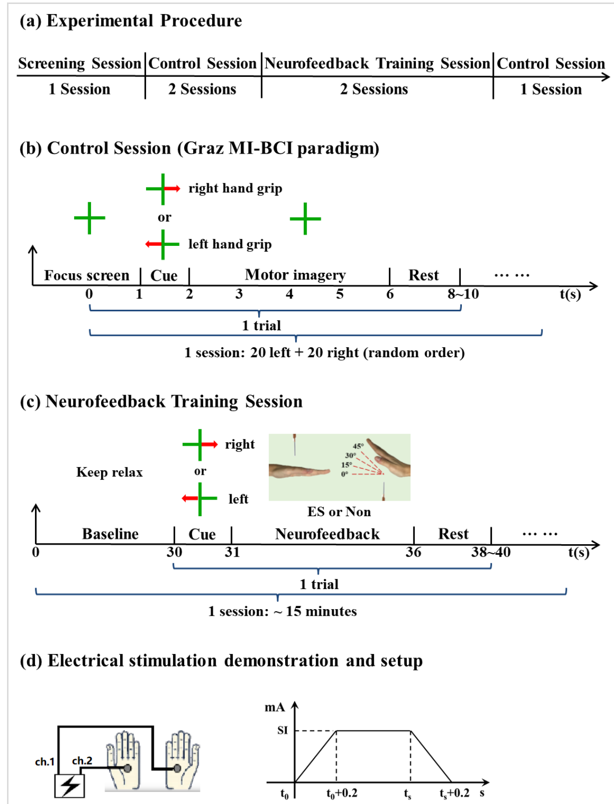
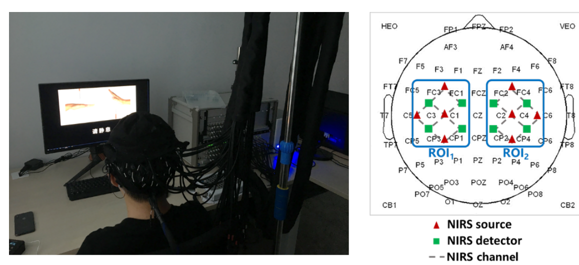


Figure 1. Experimental paradigm. (a) Experimental procedure. (b) previous and posterior control sessions using Graz MI-BCI paradigm. (c) NFT session. (d) Electrical stimulation, SI indicates stimulation intensity individually determined, t_s indicates the end time of current Neurofeedback trial or the time when the training feature parameter lower than the threshold again.



(a) (b)

Figure 2. (a) NFT experiment situation and EEG-NIRS equipment; (b) EEG-NIRS channels covered regions

$$lrERD_{C4-C3} = RP_{ERD,C4} - RP_{ERD,C3} \tag{2}$$

where the relative ERD power (RP_{ERD}) at C3 or C4 channel using the power of *n* trials across specific time period(P_n) according to equation defined as follows [11]:

$$P_{relax} = \frac{1}{T_{relax}} \sum_{n \in T_{relax}} P_n \tag{3}$$

$$RP_{ERD} = \frac{P_{task} - P_{relax}}{P_{relax}} \times 100 \tag{5}$$

where P_{relax} and P_{task} are the average power spectra during the rest period (T_{relax}) and the task period (T_{task}). Therefore, the averaged event-related power changes in a time-frequency domain could be visualized by the event-related spectral perturbation (ERSP), which provides detailed information for ERD/ERS patterns of different tasks. An ERSP of n trials was calculated according to equation defined as follows:

$$ERSP(f,t) = \frac{1}{n}\sum_{k=1}^{n}(F_k(f,t)^2) \tag{6}$$

where $F_k(f, t)$ indicates the spectral estimation at frequency f and time t for the kth trial. The ERSP (dB) was computed through short-time Fourier transform (STFT) with a 256 points Hanning-tapered window from EEGLAB [13]. For the baseline-normalized ERSP, the mean power changes in a baseline period were subtracted from each spectral estimation.

E. Integrating EEG-NIRS patterns for Classification

The common spatial pattern (CSP) and support vector machine (SVM) algorithm were applied to calculate the classification accuracies. Classification results of the control sessions (40 trials for LH-MI and 40 trials for RH-MI) were estimated using a leave-one-out paradigm. based on EEG, NIRS and EEG-NIRS pattern fusion respectively. We probed its feasibility of improving system classification performance with EEG-NIRS fusion. The methodology is described as follows:

EEG and NIRS patterns will be extracted respectively. X_i is the ith feature vector of EEG data and Y_j is the jth feature vector of NIRS data. m and n are the vector dimension respectively.

$$X_i = \{x_1, x_2, \cdots x_m\} \tag{7}$$

$$Y_j = \{y_1, y_2, \cdots y_n\} \tag{8}$$

And the fusion feature is Z.

$$Z = F(X_i, Y_i) = \{x_1, x_2, \cdots x_m, y_1, y_2, \cdots y_n\} \tag{9}$$

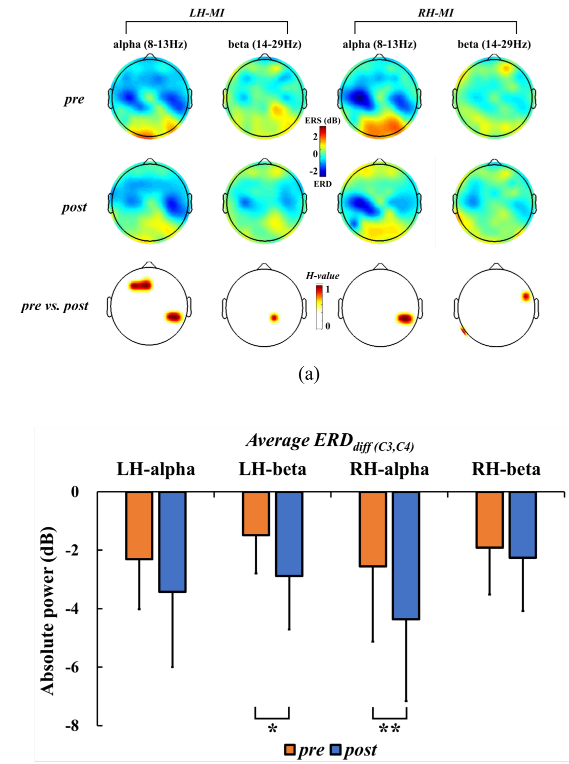
F is a simple feature fusion by splicing feature vectors. New classification accuracy is based on Z. We further calculated classification results based on SVM decision fusion method to explore the performance improvement of MI-BCI system.

III. Results And Analysis

A. ERSP patterns analysis

To investigate spatial distributions of the cortical activations for previous and posterior control sessions, the averaged topographical distribution maps of absolute ERDs at two salient frequency bands (alpha: 8-13 Hz, beta: 14-29 Hz) are also shown across all subjects in Fig. 3(a). Additionally, the H-value (0 or 1) maps of paired-samples t-test ($p < 0.01$) present significant differences at different frequency bands covering some sensorimotor cortical areas. These results indicate the contralateral dominance could be observed more clearly for posterior control sessions.

To quantitatively verify the effectiveness of the proposed NFT, this two salient ERD bands were also selected respectively to compare their absolute powers between previous and posterior control sessions. Fig. 3(b) shows the averaged absolute ERD powers of LH-MI and RH-MI task across all subjects. After Shapiro-Wilk test of normality, the paired-samples t-test yielded significant differences between



(a)

(b)

Figure 3. EEG based ERSP patterns. (a) Averaged topographical distributions of multi-band absolute ERD powers and H-value (0 or 1) of paired-samples t-test ($p < 0.01$). The deep red of H-value maps indicates visualized significant differences. (b) Comparisons of the averaged absolute ERDs before and after NFT across all subjects (Error bar indicates the SD, * represents the significant difference, *$p < 0.01$ and **$p < 0.001$).

previous and posterior control sessions at different conditions (*pre* versus *post* at beta-ERD of LH-MI: $t(15) = 2.347$, $p = 0.033$, $d = 0.59$; alpha-ERD of RH-MI: $t(15) = 3.661$, $p = 0.002$, $d = 0.92$). The quantitatively ERD patterns indicate relatively consistent trend with topographical phenomena. Therefore, the proposed visual-haptic NFT could indeed improve cortical activations significantly during left or right hand MI.

B. Classification performance

As shown in Fig. 4, we compared the 2-class MI-BCI (LH-MI *vs.* RH-MI) classification performance under different calibration methods before and after NFT cross all subjects. The classification accuracy before training is relatively low. Classification accuracy based on different data of most subjects is less than 70% and slightly higher than random level (50%), Which is related to low MI-BCI performance of certain subject. And after applying feature fusion (only CSP patterns of EEG and NIRS were spliced together), system mean classification accuracy is 76.63%. This result is significantly higher than that of single data classification (NIRS: 64.69% and EEG: 71.09%; ANOVA: $F(2, 30) = 39.845$, $p = 0.000 < 0.001$). Bonferroni post-conditional paired comparison results (*Hybrid-pre vs. NIRS-pre:* p=0.000, *Hybrid-pre vs. EEG-pre*: p=0.000) verify classification accuracy of EEG-NIRS feature fusion is significantly higher than that of single data.

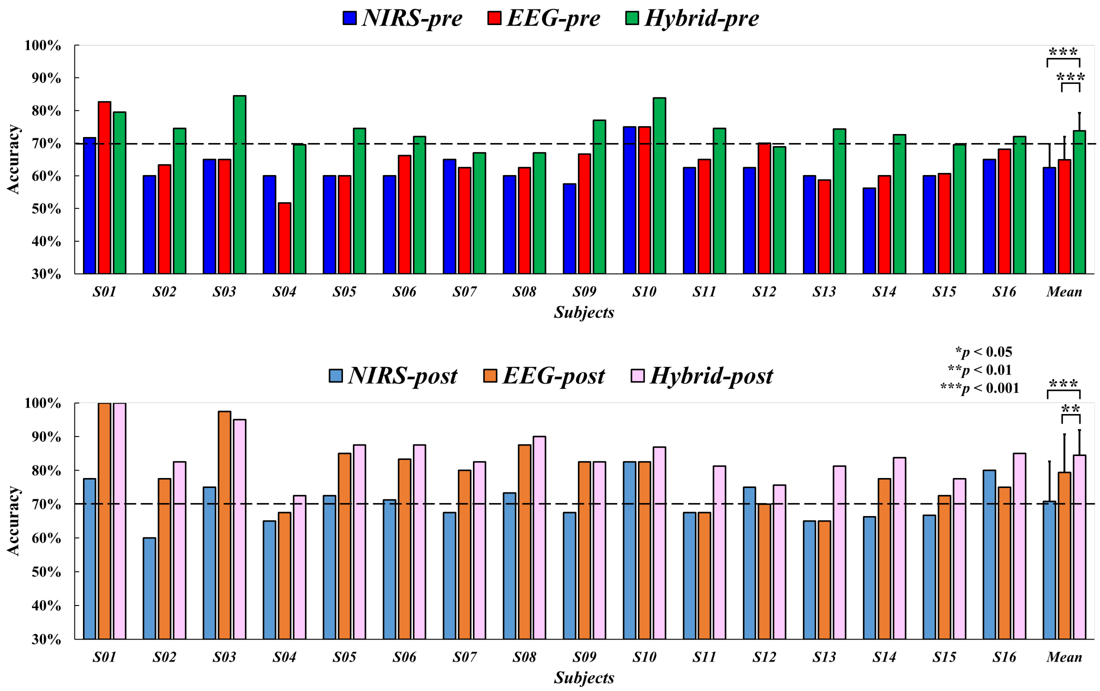


Figure 4. Comparisons of the MI-BCI performance under different calibration methods before and after NFT cross all subjects (Error bar indicates the SD, * represents the significant difference, *p < 0.01 and **p < 0.001). The black dashed line indicates 70% accuracy.

The classification accuracy after training is apparently higher than before training, which is consistent with EEG results. The classification accuracies of most subjects based on different data are more than 70%, which is because of visual-haptic neurofeedback training effect. Data fusion method makes system mean classification accuracy become 84.45%, significantly higher than single data results (NIRS: 70.78% and EEG: 79.43%; ANOVA: F (2, 30) = 17.919, p = 0.000<0.001). Bonferroni post-conditional paired comparison results *(Hybrid-pre vs. NIRS-pre*: p=0.000, *Hybrid-pre vs. EEG-pre:* p=0.000) verify classification accuracy of EEG-NIRS feature fusion is significantly higher than that of single data.

IV. Conclusion

Both multi-band ERD patterns, classification performance were significantly improved after our BCI based NFT. Additionally, the classifier constructed on integrating EEG and NIRS patterns was significantly superior to that only with independent information. These findings validate the feasibility of our proposed EEG-NIRS patterns fusion approach to improve BCI performance. And it is promising to optimize MI-BCI based NFT manner and evaluate the effects of motor training.

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